

Michael Baron

PROBABILITY AND STATISTICS FOR COMPUTER SCIENTISTS

SECOND EDITION



$$\lambda_A \pi_{i-1} = \lambda_S \pi_i$$

239

10^6



CRC Press

Taylor & Francis Group

A CHAPMAN & HALL BOOK

PROBABILITY AND STATISTICS FOR COMPUTER SCIENTISTS

SECOND EDITION

PROBABILITY AND STATISTICS FOR COMPUTER SCIENTISTS

SECOND EDITION

Michael Baron

University of Texas at Dallas
Richardson, USA



CRC Press

Taylor & Francis Group

Boca Raton London New York

CRC Press is an imprint of the
Taylor & Francis Group, an **informa** business

A CHAPMAN & HALL BOOK

CRC Press
Taylor & Francis Group
6000 Broken Sound Parkway NW, Suite 300
Boca Raton, FL 33487-2742

© 2014 by Taylor & Francis Group, LLC
CRC Press is an imprint of Taylor & Francis Group, an Informa business

No claim to original U.S. Government works
Version Date: 20130625

International Standard Book Number-13: 978-1-4822-1410-9 (eBook - ePub)

This book contains information obtained from authentic and highly regarded sources. Reasonable efforts have been made to publish reliable data and information, but the author and publisher cannot assume responsibility for the validity of all materials or the consequences of their use. The authors and publishers have attempted to trace the copyright holders of all material reproduced in this publication and apologize to copyright holders if permission to publish in this form has not been obtained. If any copyright material has not been acknowledged please write and let us know so we may rectify in any future reprint.

Except as permitted under U.S. Copyright Law, no part of this book may be reprinted, reproduced, transmitted, or utilized in any form by any electronic, mechanical, or other means, now known or hereafter invented, including photocopying, microfilming, and recording, or in any information storage or retrieval system, without written permission from the publishers.

For permission to photocopy or use material electronically from this work, please access www.copyright.com (<http://www.copyright.com/>) or contact the Copyright Clearance Center, Inc. (CCC), 222 Rosewood Drive, Danvers, MA 01923, 978-750-8400. CCC is a not-for-profit organization that provides licenses and registration for a variety of users. For organizations that have been granted a photocopy license by the CCC, a separate system of payment has been arranged.

Trademark Notice: Product or corporate names may be trademarks or registered trademarks, and are used only for identification and explanation without intent to infringe.

Visit the Taylor & Francis Web site at
<http://www.taylorandfrancis.com>

and the CRC Press Web site at
<http://www.crcpress.com>

To my parents – Genrietta and Izrael-Vulf Baron

Contents

List of Figures	xv
List of Tables	xix
Preface	xxi
1 Introduction and Overview	1
1.1 Making decisions under uncertainty	1
1.2 Overview of this book	3
Summary and conclusions	5
Exercises	5
I Probability and Random Variables	7
2 Probability	9
2.1 Events and their probabilities	9
2.1.1 Outcomes, events, and the sample space	10
2.1.2 Set operations	11
2.2 Rules of Probability	13
2.2.1 Axioms of Probability	14
2.2.2 Computing probabilities of events	15
2.2.3 Applications in reliability	18
2.3 Combinatorics	20
2.3.1 Equally likely outcomes	20
2.3.2 Permutations and combinations	22
2.4 Conditional probability and independence	27
Summary and conclusions	32
Exercises	33

3	Discrete Random Variables and Their Distributions	39
3.1	Distribution of a random variable	40
3.1.1	Main concepts	40
3.1.2	Types of random variables	43
3.2	Distribution of a random vector	44
3.2.1	Joint distribution and marginal distributions	44
3.2.2	Independence of random variables	45
3.3	Expectation and variance	47
3.3.1	Expectation	47
3.3.2	Expectation of a function	48
3.3.3	Properties	48
3.3.4	Variance and standard deviation	49
3.3.5	Covariance and correlation	51
3.3.6	Properties	52
3.3.7	Chebyshev's inequality	54
3.3.8	Application to finance	55
3.4	Families of discrete distributions	57
3.4.1	Bernoulli distribution	57
3.4.2	Binomial distribution	58
3.4.3	Geometric distribution	60
3.4.4	Negative Binomial distribution	63
3.4.5	Poisson distribution	64
3.4.6	Poisson approximation of Binomial distribution	66
	Summary and conclusions	67
	Exercises	68
4	Continuous Distributions	75
4.1	Probability density	75
4.2	Families of continuous distributions	80
4.2.1	Uniform distribution	80
4.2.2	Exponential distribution	82
4.2.3	Gamma distribution	84
4.2.4	Normal distribution	89
4.3	Central Limit Theorem	92
	Summary and conclusions	95
	Exercises	96

5	Computer Simulations and Monte Carlo Methods	101
5.1	Introduction	101
5.1.1	Applications and examples	102
5.2	Simulation of random variables	103
5.2.1	Random number generators	104
5.2.2	Discrete methods	105
5.2.3	Inverse transform method	108
5.2.4	Rejection method	110
5.2.5	Generation of random vectors	113
5.2.6	Special methods	113
5.3	Solving problems by Monte Carlo methods	114
5.3.1	Estimating probabilities	114
5.3.2	Estimating means and standard deviations	118
5.3.3	Forecasting	119
5.3.4	Estimating lengths, areas, and volumes	120
5.3.5	Monte Carlo integration	122
	Summary and conclusions	124
	Exercises	125
II	Stochastic Processes	129
6	Stochastic Processes	131
6.1	Definitions and classifications	132
6.2	Markov processes and Markov chains	133
6.2.1	Markov chains	134
6.2.2	Matrix approach	138
6.2.3	Steady-state distribution	142
6.3	Counting processes	148
6.3.1	Binomial process	148
6.3.2	Poisson process	152
6.4	Simulation of stochastic processes	157
	Summary and conclusions	160
	Exercises	160

7	Queuing Systems	167
7.1	Main components of a queuing system	168
7.2	The Little's Law	170
7.3	Bernoulli single-server queuing process	173
7.3.1	Systems with limited capacity	176
7.4	M/M/1 system	178
7.4.1	Evaluating the system's performance	181
7.5	Multiserver queuing systems	185
7.5.1	Bernoulli k -server queuing process	186
7.5.2	M/M/ k systems	189
7.5.3	Unlimited number of servers and M/M/ ∞	192
7.6	Simulation of queuing systems	193
	Summary and conclusions	197
	Exercises	198
III	Statistics	205
8	Introduction to Statistics	207
8.1	Population and sample, parameters and statistics	208
8.2	Simple descriptive statistics	211
8.2.1	Mean	211
8.2.2	Median	213
8.2.3	Quantiles, percentiles, and quartiles	217
8.2.4	Variance and standard deviation	219
8.2.5	Standard errors of estimates	221
8.2.6	Interquartile range	222
8.3	Graphical statistics	223
8.3.1	Histogram	224
8.3.2	Stem-and-leaf plot	227
8.3.3	Boxplot	229
8.3.4	Scatter plots and time plots	231
	Summary and conclusions	233
	Exercises	233

9 Statistical Inference I**237**

9.1	Parameter estimation	238
9.1.1	Method of moments	239
9.1.2	Method of maximum likelihood	242
9.1.3	Estimation of standard errors	246
9.2	Confidence intervals	247
9.2.1	Construction of confidence intervals: a general method	248
9.2.2	Confidence interval for the population mean	250
9.2.3	Confidence interval for the difference between two means	251
9.2.4	Selection of a sample size	253
9.2.5	Estimating means with a given precision	254
9.3	Unknown standard deviation	255
9.3.1	Large samples	255
9.3.2	Confidence intervals for proportions	256
9.3.3	Estimating proportions with a given precision	258
9.3.4	Small samples: Student's t distribution	259
9.3.5	Comparison of two populations with unknown variances	261
9.4	Hypothesis testing	264
9.4.1	Hypothesis and alternative	265
9.4.2	Type I and Type II errors: level of significance	266
9.4.3	Level α tests: general approach	267
9.4.4	Rejection regions and power	269
9.4.5	Standard Normal null distribution (Z-test)	270
9.4.6	Z-tests for means and proportions	272
9.4.7	Pooled sample proportion	274
9.4.8	Unknown σ : T-tests	275
9.4.9	Duality: two-sided tests and two-sided confidence intervals	277
9.4.10	P-value	280
9.5	Inference about variances	285
9.5.1	Variance estimator and Chi-square distribution	286
9.5.2	Confidence interval for the population variance	287
9.5.3	Testing variance	289
9.5.4	Comparison of two variances. F-distribution.	292
9.5.5	Confidence interval for the ratio of population variances	294
9.5.6	F-tests comparing two variances	296
	Summary and conclusions	299
	Exercises	300

10 Statistical Inference II	305
10.1 Chi-square tests	305
10.1.1 Testing a distribution	306
10.1.2 Testing a family of distributions	308
10.1.3 Testing independence	310
10.2 Nonparametric statistics	314
10.2.1 Sign test	315
10.2.2 Wilcoxon signed rank test	317
10.2.3 Mann-Whitney-Wilcoxon rank sum test	322
10.3 Bootstrap	328
10.3.1 Bootstrap distribution and all bootstrap samples	328
10.3.2 Computer generated bootstrap samples	333
10.3.3 Bootstrap confidence intervals	335
10.4 Bayesian inference	339
10.4.1 Prior and posterior	340
10.4.2 Bayesian estimation	345
10.4.3 Bayesian credible sets	347
10.4.4 Bayesian hypothesis testing	351
Summary and conclusions	352
Exercises	353
11 Regression	361
11.1 Least squares estimation	362
11.1.1 Examples	362
11.1.2 Method of least squares	364
11.1.3 Linear regression	365
11.1.4 Regression and correlation	367
11.1.5 Overfitting a model	368
11.2 Analysis of variance, prediction, and further inference	369
11.2.1 ANOVA and R-square	369
11.2.2 Tests and confidence intervals	371
11.2.3 Prediction	377
11.3 Multivariate regression	381

11.3.1	Introduction and examples	381
11.3.2	Matrix approach and least squares estimation	382
11.3.3	Analysis of variance, tests, and prediction	384
11.4	Model building	390
11.4.1	Adjusted R-square	390
11.4.2	Extra sum of squares, partial F-tests, and variable selection	391
11.4.3	Categorical predictors and dummy variables	394
	Summary and conclusions	397
	Exercises	397
IV	Appendix	403
12	Appendix	405
12.1	Inventory of distributions	405
12.1.1	Discrete families	405
12.1.2	Continuous families	407
12.2	Distribution tables	411
12.3	Calculus review	428
12.3.1	Inverse function	428
12.3.2	Limits and continuity	428
12.3.3	Sequences and series	429
12.3.4	Derivatives, minimum, and maximum	429
12.3.5	Integrals	431
12.4	Matrices and linear systems	434
12.5	Answers to selected exercises	439
	Index	445

Chapter 3

Discrete Random Variables and Their Distributions

3.1	Distribution of a random variable	40
3.1.1	Main concepts	40
3.1.2	Types of random variables	43
3.2	Distribution of a random vector	44
3.2.1	Joint distribution and marginal distributions	44
3.2.2	Independence of random variables	45
3.3	Expectation and variance	47
3.3.1	Expectation	47
3.3.2	Expectation of a function	48
3.3.3	Properties	48
3.3.4	Variance and standard deviation	49
3.3.5	Covariance and correlation	51
3.3.6	Properties	52
3.3.7	Chebyshev's inequality	54
3.3.8	Application to finance	55
3.4	Families of discrete distributions	57
3.4.1	Bernoulli distribution	57
3.4.2	Binomial distribution	58
3.4.3	Geometric distribution	60
3.4.4	Negative Binomial distribution	63
3.4.5	Poisson distribution	64
3.4.6	Poisson approximation of Binomial distribution	66
	Summary and conclusions	67
	Exercises	68

This chapter introduces the concept of a random variable and studies discrete distributions in detail. Continuous distributions are discussed in Chapter 4.

3.1 Distribution of a random variable

3.1.1 Main concepts

DEFINITION 3.1

A **random variable** is a function of an outcome,

$$X = f(\omega).$$

In other words, it is a quantity that depends on chance.

The domain of a random variable is the sample space Ω . Its range can be the set of all real numbers $\mathbb{R}$, or only the positive numbers $(0, +\infty)$, or the integers $\mathbb{Z}$, or the interval $(0, 1)$, etc., depending on what possible values the random variable can potentially take.

Once an experiment is completed, and the outcome ω is known, the value of random variable $X(\omega)$ becomes determined.

Example 3.1. Consider an experiment of tossing 3 fair coins and counting the number of heads. Certainly, the same model suits the number of girls in a family with 3 children, the number of 1's in a random binary code consisting of 3 characters, etc.

Let X be the number of heads (girls, 1's). Prior to an experiment, its value is not known. All we can say is that X has to be an integer between 0 and 3. Since assuming each value is an event, we can compute probabilities,

$$\begin{aligned} P\{X = 0\} &= P\{\text{three tails}\} = P\{TTT\} = \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) \left(\frac{1}{2}\right) = \frac{1}{8} \\ P\{X = 1\} &= P\{HTT\} + P\{THT\} + P\{TTH\} = \frac{3}{8} \\ P\{X = 2\} &= P\{HHT\} + P\{HTH\} + P\{THH\} = \frac{3}{8} \\ P\{X = 3\} &= P\{HHH\} = \frac{1}{8} \end{aligned}$$

Summarizing,

x	$P\{X = x\}$
0	1/8
1	3/8
2	3/8
3	1/8
Total	1

◇

This table contains everything that is known about random variable X prior to the experiment. Before we know the outcome ω , we cannot tell what X equals to. However, we can list all the possible values of X and determine the corresponding probabilities.

DEFINITION 3.2

Collection of all the probabilities related to X is the **distribution** of X . The function

$$P(x) = \mathbf{P}\{X = x\}$$

is the **probability mass function**, or **pmf**. The **cumulative distribution function**, or **cdf** is defined as

$$F(x) = \mathbf{P}\{X \leq x\} = \sum_{y \leq x} \mathbf{P}(y). \quad (3.1)$$

The set of possible values of X is called the **support** of the distribution F .

For every outcome ω , the variable X takes one and only one value x . This makes events $\{X = x\}$ disjoint and exhaustive, and therefore,

$$\sum_x P(x) = \sum_x \mathbf{P}\{X = x\} = 1.$$

Looking at (3.1), we can conclude that the cdf $F(x)$ is a non-decreasing function of x , always between 0 and 1, with

$$\lim_{x \downarrow -\infty} F(x) = 0 \quad \text{and} \quad \lim_{x \uparrow +\infty} F(x) = 1.$$

Between any two subsequent values of X , $F(x)$ is constant. It jumps by $P(x)$ at each possible value x of X (see Fig. 3.1, right).

Recall that one way to compute the probability of an event is to add probabilities of all the outcomes in it. Hence, for any set A ,

$$\mathbf{P}\{X \in A\} = \sum_{x \in A} P(x).$$

When A is an interval, its probability can be computed directly from the cdf $F(x)$,

$$\mathbf{P}\{a < X \leq b\} = F(b) - F(a).$$

Example 3.2. The pmf and cdf of X in Example 3.1 are shown in Fig. 3.1.

◇

MATLAB DEMO. The following MATLAB code simulates 3 coin tosses and computes a value of X 10,000 times.

```
N = 10000;           % Number of simulations
U = rand(3,N);       % a 3-by-N matrix of random numbers from [0,1]
Y = (U < 0.5);        % Y=1 (heads) if U < 0.5, otherwise Y=0 (tails)
X = sum(Y);           % Sums across columns. X = number of heads
hist(X);              % Histogram of X
```

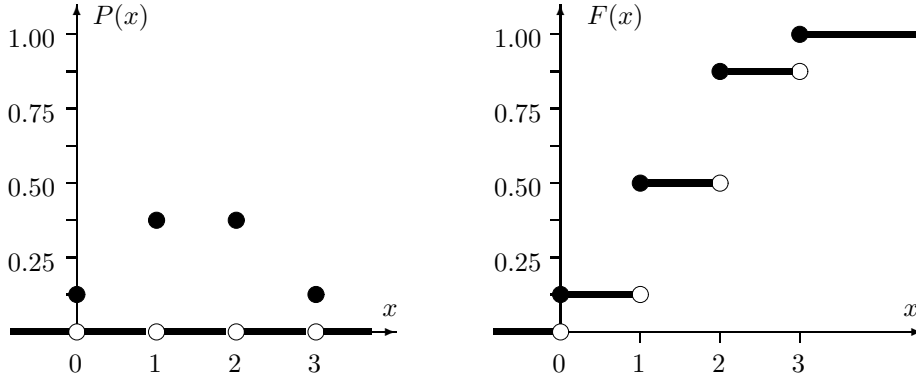


FIGURE 3.1: The probability mass function $P(x)$ and the cumulative distribution function $F(x)$ for Example 3.1. White circles denote excluded points.

On the obtained histogram, the two middle columns for $X = 1$ and $X = 2$ are about 3 times higher than the columns on each side, for $X = 0$ and $X = 3$. That is, in a run of 10,000 simulations, values 1 and 2 are attained three times more often than 0 and 3. This agrees with the pmf $P(0) = P(3) = 1/8$, $P(1) = P(2) = 3/8$.

Example 3.3 (ERRORS IN INDEPENDENT MODULES). A program consists of two modules. The number of errors X_1 in the first module has the pmf $P_1(x)$, and the number of errors X_2 in the second module has the pmf $P_2(x)$, independently of X_1 , where

x	$P_1(x)$	$P_2(x)$
0	0.5	0.7
1	0.3	0.2
2	0.1	0.1
3	0.1	0

Find the pmf and cdf of $Y = X_1 + X_2$, the total number of errors.

Solution. We break the problem into steps. First, determine all possible values of Y , then compute the probability of each value. Clearly, the number of errors Y is an integer that can be as low as $0 + 0 = 0$ and as high as $3 + 2 = 5$. Since $P_2(3) = 0$, the second module has at most 2 errors. Next,

$$\begin{aligned}
 P_Y(0) &= P\{Y = 0\} = \mathbf{P}\{X_1 = X_2 = 0\} = P_1(0)P_2(0) \\
 &= (0.5)(0.7) = 0.35 \\
 P_Y(1) &= P\{Y = 1\} = P_1(0)P_2(1) + P_1(1)P_2(0) \\
 &= (0.5)(0.2) + (0.3)(0.7) = 0.31 \\
 P_Y(2) &= P\{Y = 2\} = P_1(0)P_2(2) + P_1(1)P_2(1) + P_1(2)P_2(0) \\
 &= (0.5)(0.1) + (0.3)(0.2) + (0.1)(0.7) = 0.18 \\
 P_Y(3) &= P\{Y = 3\} = P_1(1)P_2(2) + P_1(2)P_2(1) + P_1(3)P_2(0) \\
 &= (0.3)(0.1) + (0.1)(0.2) + (0.1)(0.7) = 0.12 \\
 P_Y(4) &= P\{Y = 4\} = P_1(2)P_2(2) + P_1(3)P_2(1) \\
 &= (0.1)(0.1) + (0.1)(0.2) = 0.03 \\
 P_Y(5) &= P\{Y = 5\} = P_1(3)P_2(2) = (0.1)(0.1) = 0.01
 \end{aligned}$$

Now check:

$$\sum_{y=0}^5 P_Y(y) = 0.35 + 0.31 + 0.18 + 0.12 + 0.03 + 0.01 = 1,$$

thus we *probably* counted all the possibilities and did not miss any (we just wanted to emphasize that simply getting $\sum P(x) = 1$ does not guarantee that we made no mistake in our solution. However, if this equality is not satisfied, we have a mistake for sure).

The cumulative function can be computed as

$$\begin{aligned} F_Y(0) &= P_Y(0) = 0.35 \\ F_Y(1) &= F_Y(0) + P_Y(1) = 0.35 + 0.31 = 0.66 \\ F_Y(2) &= F_Y(1) + P_Y(2) = 0.66 + 0.18 = 0.84 \\ F_Y(3) &= F_Y(2) + P_Y(3) = 0.84 + 0.12 = 0.96 \\ F_Y(4) &= F_Y(3) + P_Y(4) = 0.96 + 0.03 = 0.99 \\ F_Y(5) &= F_Y(4) + P_Y(5) = 0.99 + 0.01 = 1.00 \end{aligned}$$

Between the values of Y , $F(x)$ is constant. ◇

3.1.2 Types of random variables

So far, we are dealing with *discrete random variables*. These are variables whose range is finite or countable. In particular, it means that their values can be listed, or arranged in a sequence. Examples include the number of jobs submitted to a printer, the number of errors, the number of error-free modules, the number of failed components, and so on. Discrete variables don't have to be integers. For example, the *proportion* of defective components in a lot of 100 can be 0, 1/100, 2/100, ..., 99/100, or 1. This variable assumes 101 different values, so it is discrete, although not an integer.

On the contrary, *continuous random variables* assume a whole interval of values. This could be a bounded interval (a, b) , or an unbounded interval such as $(a, +\infty)$, $(-\infty, b)$, or $(-\infty, +\infty)$. Sometimes, it may be a union of several such intervals. Intervals are uncountable, therefore, all values of a random variable cannot be listed in this case. Examples of continuous variables include various times (software installation time, code execution time, connection time, waiting time, lifetime), also physical variables like weight, height, voltage, temperature, distance, the number of miles per gallon, etc. We shall discuss continuous random variables in detail in Chapter 4.

Example 3.4. For comparison, observe that a long jump is formally a continuous random variable because an athlete can jump any distance within some range. Results of a high jump, however, are discrete because the bar can only be placed on a finite number of heights. ◇

Notice that rounding a continuous random variable, say, to the nearest integer makes it discrete.

Sometimes we can see *mixed random variables* that are discrete on some range of values and continuous elsewhere. For example, see a remark about waiting times on page 182.

Example 3.5. A job is sent to a printer. Let X be the waiting time before the job starts printing. With some probability, this job appears first in line and starts printing immediately, $X = 0$. It is also possible that the job is first in line but it takes 20 seconds for the printer to warm up, in which case $X = 20$. So far, the variable has a discrete behavior with a positive pmf $P(x)$ at $x = 0$ and $x = 20$. However, if there are other jobs in a queue, then X depends on the time it takes to print them, which is a continuous random variable. Using a popular jargon, besides “point masses” at $x = 0$ and $x = 20$, the variable is continuous, taking values in $(0, +\infty)$. Thus, X is neither discrete nor continuous. It is mixed. $\diamond$

3.2 Distribution of a random vector

Often we deal with several random variables simultaneously. We may look at the size of a RAM and the speed of a CPU, the price of a computer and its capacity, temperature and humidity, technical and artistic performance, etc.

3.2.1 Joint distribution and marginal distributions

DEFINITION 3.3

If X and Y are random variables, then the pair (X, Y) is a **random vector**. Its distribution is called the **joint distribution** of X and Y . Individual distributions of X and Y are then called the **marginal distributions**.

Although we talk about two random variables in this section, all the concepts extend to a vector $(X_1, X_2, \dots, X_n)$ of n components and its joint distribution.

Similarly to a single variable, the *joint distribution* of a vector is a collection of probabilities for a vector (X, Y) to take a value (x, y) . Recall that two vectors are equal,

$$(X, Y) = (x, y),$$

if $X = x$ and $Y = y$. This “and” means the intersection, therefore, the *joint probability mass function* of X and Y is

$$P(x, y) = \mathbf{P}\{(X, Y) = (x, y)\} = \mathbf{P}\{X = x \cap Y = y\}.$$

Again, $\{(X, Y) = (x, y)\}$ are exhaustive and mutually exclusive events for different pairs (x, y) , therefore,

$$\sum_x \sum_y P(x, y) = 1.$$

The joint distribution of (X, Y) carries the complete information about the behavior of this random vector. In particular, the marginal probability mass functions of X and Y can be obtained from the joint pmf by the Addition Rule.

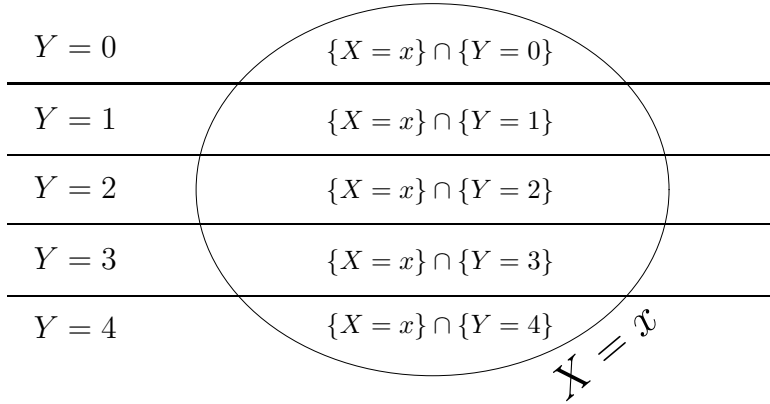


FIGURE 3.2: Addition Rule: computing marginal probabilities from the joint distribution.

Addition Rule

$$\begin{aligned}
 P_X(x) &= \mathbf{P}\{X = x\} = \sum_y P_{(X,Y)}(x, y) \\
 P_Y(y) &= \mathbf{P}\{Y = y\} = \sum_x P_{(X,Y)}(x, y)
 \end{aligned}
 \tag{3.2}$$

That is, to get the marginal pmf of one variable, we add the joint probabilities over all values of the other variable.

The Addition Rule is illustrated in Figure 3.2. Events $\{Y = y\}$ for different values of y partition the sample space Ω . Hence, their intersections with $\{X = x\}$ partition the event $\{X = x\}$ into mutually exclusive parts. By the rule for the union of mutually exclusive events, formula (2.4) on p. 16, their probabilities should be added. These probabilities are precisely $P_{(X,Y)}(x, y)$.

In general, the joint distribution cannot be computed from marginal distributions because they carry no information about interrelations between random variables. For example, marginal distributions cannot tell whether variables X and Y are independent or dependent.

3.2.2 Independence of random variables

DEFINITION 3.4 —

Random variables X and Y are **independent** if

$$P_{(X,Y)}(x, y) = P_X(x)P_Y(y)$$

for all values of x and y . This means, events $\{X = x\}$ and $\{Y = y\}$ are independent for all x and y ; in other words, variables X and Y take their values independently of each other.

In problems, to show independence of X and Y , we have to check whether the joint pmf factors into the product of marginal pmfs for *all* pairs x and y . To prove dependence, we only need to present one counterexample, a pair (x, y) with $P(x, y) \neq P_X(x)P_Y(y)$.

Example 3.6. A program consists of two modules. The number of errors, X , in the first module and the number of errors, Y , in the second module have the joint distribution, $P(0, 0) = P(0, 1) = P(1, 0) = 0.2$, $P(1, 1) = P(1, 2) = P(1, 3) = 0.1$, $P(0, 2) = P(0, 3) = 0.05$. Find (a) the marginal distributions of X and Y , (b) the probability of no errors in the first module, and (c) the distribution of the total number of errors in the program. Also, (d) find out if errors in the two modules occur independently.

Solution. It is convenient to organize the joint pmf of X and Y in a table. Adding rowwise and columnwise, we get the marginal pmfs,

$P_{(X,Y)}(x, y)$		y				$P_X(x)$
		0	1	2	3	
x	0	0.20	0.20	0.05	0.05	0.50
	1	0.20	0.10	0.10	0.10	0.50
$P_Y(y)$		0.40	0.30	0.15	0.15	1.00

This solves (a).

(b) $P_X(0) = 0.50$.

(c) Let $Z = X + Y$ be the total number of errors. To find the distribution of Z , we first identify its possible values, then find the probability of each value. We see that Z can be as small as 0 and as large as 4. Then,

$$\begin{aligned}
 P_Z(0) &= \mathbf{P}\{X + Y = 0\} = \mathbf{P}\{X = 0 \cap Y = 0\} = P(0, 0) = 0.20, \\
 P_Z(1) &= \mathbf{P}\{X = 0 \cap Y = 1\} + \mathbf{P}\{X = 1 \cap Y = 0\} \\
 &= P(0, 1) + P(1, 0) = 0.20 + 0.20 = 0.40, \\
 P_Z(2) &= P(0, 2) + P(1, 1) = 0.05 + 0.10 = 0.15, \\
 P_Z(3) &= P(0, 3) + P(1, 2) = 0.05 + 0.10 = 0.15, \\
 P_Z(4) &= P(1, 3) = 0.10.
 \end{aligned}$$

It is a good check to verify that $\sum_z P_Z(z) = 1$.

(d) To decide on the independence of X and Y , check if their joint pmf factors into a product of marginal pmfs. We see that $P_{(X,Y)}(0, 0) = 0.2$ indeed equals $P_X(0)P_Y(0) = (0.5)(0.4)$. Keep checking... Next, $P_{(X,Y)}(0, 1) = 0.2$ whereas $P_X(0)P_Y(1) = (0.5)(0.3) = 0.15$. There is no need to check further. We found a pair of x and y that violates the formula for independent random variables. Therefore, the numbers of errors in two modules are dependent. $\diamond$

3.3 Expectation and variance

The distribution of a random variable or a random vector, the full collection of related probabilities, contains the entire information about its behavior. This detailed information can be summarized in a few vital characteristics describing the average value, the most likely value of a random variable, its spread, variability, etc. The most commonly used are the *expectation*, *variance*, *standard deviation*, *covariance*, and *correlation*, introduced in this section. Also rather popular and useful are the *mode*, *moments*, *quantiles*, and *interquartile range* that we discuss in Sections 8.1 and 9.1.1.

3.3.1 Expectation

DEFINITION 3.5

Expectation or **expected value** of a random variable X is its mean, the average value.

We know that X can take different values with different probabilities. For this reason, its average value is *not* just the average of all its values. Rather, it is a *weighted average*.

Example 3.7. Consider a variable that takes values 0 and 1 with probabilities $P(0) = P(1) = 0.5$. That is,

$$X = \begin{cases} 0 & \text{with probability } 1/2 \\ 1 & \text{with probability } 1/2 \end{cases}$$

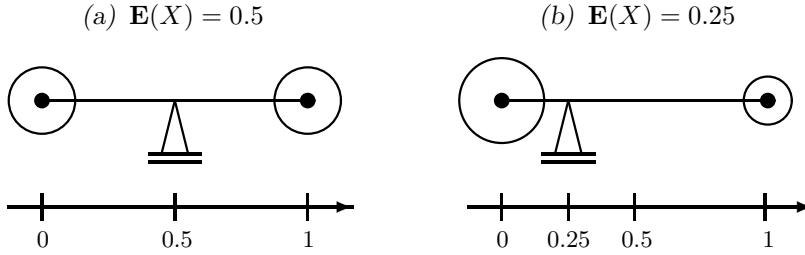
Observing this variable many times, we shall see $X = 0$ about 50% of times and $X = 1$ about 50% of times. The average value of X will then be close to 0.5, so it is reasonable to have $\mathbf{E}(X) = 0.5$. $\diamond$

Now let's see how unequal probabilities of $X = 0$ and $X = 1$ affect the expectation $\mathbf{E}(X)$.

Example 3.8. Suppose that $P(0) = 0.75$ and $P(1) = 0.25$. Then, in a long run, X is equal 1 only 1/4 of times, otherwise it equals 0. Suppose we earn \$1 every time we see $X = 1$. On the average, we earn \$1 every four times, or \$0.25 per each observation. Therefore, in this case $\mathbf{E}(X) = 0.25$. $\diamond$

A physical model for these two examples is shown in Figure 3.3. In Figure 3.3a, we put two equal masses, 0.5 units each, at points 0 and 1 and connect them with a firm but weightless rod. The masses represent probabilities $P(0)$ and $P(1)$. Now we look for a point at which the system will be balanced. It is symmetric, hence its balance point, *the center of gravity*, is in the middle, 0.5.

Figure 3.3b represents a model for the second example. Here the masses at 0 and 1 equal 0.75 and 0.25 units, respectively, according to $P(0)$ and $P(1)$. This system is balanced at 0.25, which is also its center of gravity.

FIGURE 3.3: *Expectation as a center of gravity.*

Similar arguments can be used to derive the general formula for the expectation.

**Expectation,
discrete case**

$$\mu = \mathbf{E}(X) = \sum_x xP(x)$$

(3.3)

This formula returns the center of gravity for a system with masses $P(x)$ allocated at points x . Expected value is often denoted by a Greek letter μ .

In a certain sense, expectation is the best forecast of X . The variable itself is random. It takes different values with different probabilities $P(x)$. At the same time, it has just one expectation $\mathbf{E}(X)$ which is non-random.

3.3.2 Expectation of a function

Often we are interested in another variable, Y , that is a function of X . For example, downloading time depends on the connection speed, profit of a computer store depends on the number of computers sold, and bonus of its manager depends on this profit. Expectation of $Y = g(X)$ is computed by a similar formula,

$$\mathbf{E}\{g(X)\} = \sum_x g(x)P(x). \quad (3.4)$$

Remark: Indeed, if g is a one-to-one function, then Y takes each value $y = g(x)$ with probability $P(x)$, and the formula for $\mathbf{E}(Y)$ can be applied directly. If g is not one-to-one, then some values of $g(x)$ will be repeated in (3.4). However, they are still multiplied by the corresponding probabilities. When we add in (3.4), these probabilities are also added, thus each value of $g(x)$ is still multiplied by the probability $P_Y(g(x))$.

3.3.3 Properties

The following *linear* properties of expectations follow directly from (3.3) and (3.4). For *any* random variables X and Y and any non-random numbers a , b , and c , we have

**Properties
of
expectations**

$$\mathbf{E}(aX + bY + c) = a \mathbf{E}(X) + b \mathbf{E}(Y) + c$$

In particular,

$$\mathbf{E}(X + Y) = \mathbf{E}(X) + \mathbf{E}(Y)$$

$$\mathbf{E}(aX) = a \mathbf{E}(X)$$

$$\mathbf{E}(c) = c$$

For **independent** X and Y ,

$$\mathbf{E}(XY) = \mathbf{E}(X) \mathbf{E}(Y)$$

(3.5)

PROOF: The first property follows from the Addition Rule (3.2). For any X and Y ,

$$\begin{aligned} \mathbf{E}(aX + bY + c) &= \sum_x \sum_y (ax + by + c) P_{(X,Y)}(x, y) \\ &= \sum_x ax \sum_y P_{(X,Y)}(x, y) + \sum_y by \sum_x P_{(X,Y)}(x, y) + c \sum_x \sum_y P_{(X,Y)}(x, y) \\ &= a \sum_x x P_X(x) + b \sum_y y P_Y(y) + c. \end{aligned}$$

The next three equalities are special cases. To prove the last property, we recall that $P_{(X,Y)}(x, y) = P_X(x)P_Y(y)$ for independent X and Y , and therefore,

$$\mathbf{E}(XY) = \sum_x \sum_y (xy) P_X(x) P_Y(y) = \sum_x x P_X(x) \sum_y y P_Y(y) = \mathbf{E}(X) \mathbf{E}(Y). \quad \square$$

Remark: The last property in (3.5) holds for some dependent variables too, hence it cannot be used to verify independence of X and Y .

Example 3.9. In Example 3.6 on p. 46,

$$\mathbf{E}(X) = (0)(0.5) + (1)(0.5) = 0.5 \text{ and}$$

$$\mathbf{E}(Y) = (0)(0.4) + (1)(0.3) + (2)(0.15) + (3)(0.15) = 1.05,$$

therefore, the expected total number of errors is

$$\mathbf{E}(X + Y) = 0.5 + 1.05 = 1.65.$$

◇

Remark: Clearly, the program will never have 1.65 errors, because the number of errors is always integer. Then, should we round 1.65 to 2 errors? Absolutely not, it would be a mistake. Although both X and Y are integers, their expectations, or average values, do not have to be integers at all.

3.3.4 Variance and standard deviation

Expectation shows where the average value of a random variable is located, or where the variable is *expected* to be, plus or minus some error. How large could this “error” be, and how much can a variable *vary* around its expectation? Let us introduce some measures of variability.

Example 3.10. Here is a rather artificial but illustrative scenario. Consider two users. One receives either 48 or 52 e-mail messages per day, with a 50-50% chance of each. The other receives either 0 or 100 e-mails, also with a 50-50% chance. What is a common feature of these two distributions, and how are they different?

We see that both users receive the same average number of e-mails:

$$\mathbf{E}(X) = \mathbf{E}(Y) = 50.$$

However, in the first case, the actual number of e-mails is always close to 50, whereas it always differs from it by 50 in the second case. The first random variable, X , is more stable; it has *low variability*. The second variable, Y , has *high variability*. $\diamond$

This example shows that variability of a random variable is measured by its distance from the mean $\mu = \mathbf{E}(X)$. In its turn, this distance is random too, and therefore, cannot serve as a characteristic of a distribution. It remains to square it and take the expectation of the result.

DEFINITION 3.6

Variance of a random variable is defined as the expected squared deviation from the mean. For discrete random variables, variance is

$$\sigma^2 = \text{Var}(X) = \mathbf{E}(X - \mathbf{E}X)^2 = \sum_x (x - \mu)^2 P(x)$$

Remark: Notice that if the distance to the mean is not squared, then the result is always $\mu - \mu = 0$ bearing no information about the distribution of X .

According to this definition, variance is always non-negative. Further, it equals 0 only if $x = \mu$ for all values of x , i.e., when X is constantly equal to μ . Certainly, a constant (non-random) variable has zero variability.

Variance can also be computed as

$$\text{Var}(X) = \mathbf{E}(X^2) - \mu^2. \quad (3.6)$$

A proof of this is left as Exercise 3.38a.

DEFINITION 3.7

Standard deviation is a square root of variance,

$$\sigma = \text{Std}(X) = \sqrt{\text{Var}(X)}$$

Continuing the Greek-letter tradition, variance is often denoted by σ^2 . Then, standard deviation is σ .

If X is measured in some units, then its mean μ has the same measurement unit as X . Variance σ^2 is measured in *squared units*, and therefore, it cannot be compared with X or μ . No matter how funny it sounds, it is rather normal to measure variance of profit in *squared dollars*, variance of class enrollment in *squared students*, and variance of available disk space in *squared gigabytes*. When a squared root is taken, the resulting standard deviation σ is again measured in the same units as X . This is the main reason of introducing yet another measure of variability, σ .

3.3.5 Covariance and correlation

Expectation, variance, and standard deviation characterize the distribution of a single random variable. Now we introduce measures of *association* of two random variables.

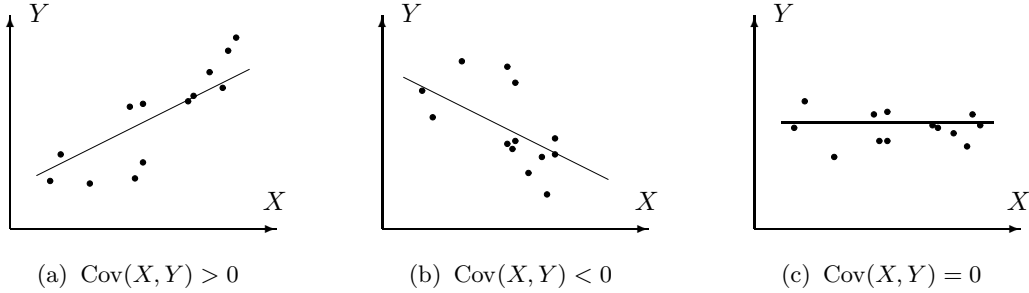


FIGURE 3.4: Positive, negative, and zero covariance.

DEFINITION 3.8

Covariance $\sigma_{XY} = \text{Cov}(X, Y)$ is defined as

$$\begin{aligned}\text{Cov}(X, Y) &= \mathbf{E}\{(X - \mathbf{E}X)(Y - \mathbf{E}Y)\} \\ &= \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y)\end{aligned}$$

It summarizes interrelation of two random variables.

Covariance is the expected product of deviations of X and Y from their respective expectations. If $\text{Cov}(X, Y) > 0$, then positive deviations $(X - \mathbf{E}X)$ are more likely to be multiplied by positive $(Y - \mathbf{E}Y)$, and negative $(X - \mathbf{E}X)$ are more likely to be multiplied by negative $(Y - \mathbf{E}Y)$. In short, large X imply large Y , and small X imply small Y . These variables are *positively correlated*, Figure 3.4a.

Conversely, $\text{Cov}(X, Y) < 0$ means that large X generally correspond to small Y and small X correspond to large Y . These variables are *negatively correlated*, Figure 3.4b.

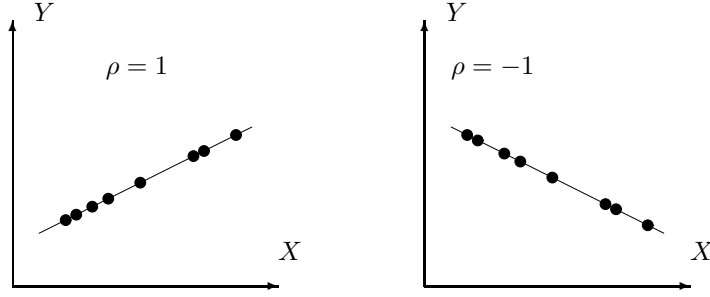
If $\text{Cov}(X, Y) = 0$, we say that X and Y are *uncorrelated*, Figure 3.4c.

DEFINITION 3.9

Correlation coefficient between variables X and Y is defined as

$$\rho = \frac{\text{Cov}(X, Y)}{(\text{Std}X)(\text{Std}Y)}$$

Correlation coefficient is a rescaled, normalized covariance. Notice that covariance $\text{Cov}(X, Y)$ has a measurement unit. It is measured in units of X multiplied by units of Y . As a result, it is not clear from its value whether X and Y are strongly or weakly correlated. Really, one has to compare $\text{Cov}(X, Y)$ with the magnitude of X and Y . Correlation coefficient performs such a comparison, and as a result, it is dimensionless.

FIGURE 3.5: Perfect correlation: $\rho = \pm 1$.

How do we interpret the value of ρ ? What possible values can it take?

As a special case of famous *Cauchy-Schwarz inequality*,

$$-1 \leq \rho \leq 1,$$

where $|\rho| = 1$ is possible only when all values of X and Y lie on a straight line, as in Figure 3.5. Further, values of ρ near 1 indicate strong positive correlation, values near (-1) show strong negative correlation, and values near 0 show weak correlation or no correlation.

3.3.6 Properties

The following properties of variances, covariances, and correlation coefficients hold for any random variables X, Y, Z , and W and any non-random numbers a, b, c and d .

Properties of variances and covariances

$$\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

$$\begin{aligned} \text{Cov}(aX + bY, cZ + dW) \\ = ac \text{Cov}(X, Z) + ad \text{Cov}(X, W) + bc \text{Cov}(Y, Z) + bd \text{Cov}(Y, W) \end{aligned}$$

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

$$\rho(X, Y) = \rho(Y, X)$$

In particular,

$$\text{Var}(aX + b) = a^2 \text{Var}(X)$$

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

$$\rho(aX + b, cY + d) = \rho(X, Y)$$

For independent X and Y ,

$$\text{Cov}(X, Y) = 0$$

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

(3.7)

PROOF: To prove the first two formulas, we only need to multiply parentheses in the definitions of variance and covariance and apply (3.5):

$$\begin{aligned}
 \text{Var}(aX + bY + c) &= \mathbf{E} \{aX + bY + c - \mathbf{E}(aX + bY + c)\}^2 \\
 &= \mathbf{E} \{(aX - a\mathbf{E}X) + (bY - b\mathbf{E}Y) + (c - c)\}^2 \\
 &= \mathbf{E} \{a(X - \mathbf{E}X)\}^2 + \mathbf{E} \{b(Y - \mathbf{E}Y)\}^2 + \mathbf{E} \{a(X - \mathbf{E}X)b(Y - \mathbf{E}Y)\} \\
 &\quad + \mathbf{E} \{b(Y - \mathbf{E}Y)a(X - \mathbf{E}X)\} \\
 &= a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).
 \end{aligned}$$

A formula for $\text{Cov}(aX + bY, cZ + dW)$ is proved similarly and is left as an exercise.

For *independent* X and Y , we have $\mathbf{E}(XY) = \mathbf{E}(X)\mathbf{E}(Y)$ from (3.5). Then, according to the definition of covariance, $\text{Cov}(X, Y) = 0$.

The other formulas follow directly from general cases, and their proofs are omitted. $\square$

We see that *independent* variables are always *uncorrelated*. The reverse is not always true. There exist some variables that are uncorrelated but not independent.

Notice that adding a constant does not affect the variables' variance or covariance. It *shifts* the whole distribution of X without changing its variability or degree of dependence of another variable. The correlation coefficient does not change even when multiplied by a constant because it is recomputed to the unit scale, due to $\text{Std}(X)$ and $\text{Std}(Y)$ in the denominator in Definition 3.9.

Example 3.11. Continuing Example 3.6, we compute

x	$P_X(x)$	$xP_X(x)$	$x - \mathbf{E}X$	$(x - \mathbf{E}X)^2 P_X(x)$
0	0.5	0	-0.5	0.125
1	0.5	0.5	0.5	0.125
$\mu_X = 0.5$			$\sigma_X^2 = 0.25$	

and (using the second method of computing variances)

y	$P_Y(y)$	$yP_Y(y)$	y^2	$y^2 P_Y(y)$
0	0.4	0	0	0
1	0.3	0.3	1	0.3
2	0.15	0.3	4	0.6
3	0.15	0.45	9	1.35
$\mu_Y = 1.05$			$\mathbf{E}(Y^2) = 2.25$	

Result: $\text{Var}(X) = 0.25$, $\text{Var}(Y) = 2.25 - 1.05^2 = 1.1475$, $\text{Std}(X) = \sqrt{0.25} = 0.5$, and $\text{Std}(Y) = \sqrt{1.1475} = 1.0712$.

Also,

$$\mathbf{E}(XY) = \sum_x \sum_y xyP(x, y) = (1)(1)(0.1) + (1)(2)(0.1) + (1)(3)(0.1) = 0.6$$

(the other five terms in this sum are 0). Therefore,

$$\text{Cov}(X, Y) = \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y) = 0.6 - (0.5)(1.05) = 0.075$$

and

$$\rho = \frac{\text{Cov}(X, Y)}{(\text{Std}X)(\text{Std}Y)} = \frac{0.075}{(0.5)(1.0712)} = 0.1400.$$

Thus, the numbers of errors in two modules are *positively and not very strongly correlated*.
 $\diamond$

<u>NOTATION</u>	μ or $\mathbf{E}(X)$	=	expectation
	σ_X^2 or $\text{Var}(X)$	=	variance
	σ_X or $\text{Std}(X)$	=	standard deviation
	σ_{XY} or $\text{Cov}(X, Y)$	=	covariance
	ρ_{XY}	=	correlation coefficient

3.3.7 Chebyshev's inequality

Knowing just the expectation and variance, one can find the range of values most likely taken by this variable. Russian mathematician *Pafnuty Chebyshev* (1821–1894) showed that any random variable X with expectation $\mu = \mathbf{E}(X)$ and variance $\sigma^2 = \text{Var}(X)$ belongs to the interval $\mu \pm \varepsilon = [\mu - \varepsilon, \mu + \varepsilon]$ with probability of at least $1 - (\sigma/\varepsilon)^2$. That is,

**Chebyshev's
inequality**

$$\mathbf{P}\{|X - \mu| > \varepsilon\} \leq \left(\frac{\sigma}{\varepsilon}\right)^2$$

for any distribution with expectation μ
and variance σ^2 and for any positive ε .

(3.8)

PROOF: Here we consider discrete random variables. For other types, the proof is similar. According to Definition 3.6,

$$\begin{aligned} \sigma^2 &= \sum_{\text{all } x} (x - \mu)^2 P(x) \geq \sum_{\text{only } x: |x - \mu| > \varepsilon} (x - \mu)^2 P(x) \\ &\geq \sum_{x: |x - \mu| > \varepsilon} \varepsilon^2 P(x) = \varepsilon^2 \sum_{x: |x - \mu| > \varepsilon} P(x) = \varepsilon^2 \mathbf{P}\{|x - \mu| > \varepsilon\}. \end{aligned}$$

Hence, $\mathbf{P}\{|x - \mu| > \varepsilon\} \leq \varepsilon^2 / \sigma^2$. $\square$

Chebyshev's inequality shows that only a large variance may allow a variable X to differ significantly from its expectation μ . In this case, the *risk* of seeing an extremely low or extremely high value of X increases. For this reason, risk is often measured in terms of a variance or standard deviation.

Example 3.12. Suppose the number of errors in a new software has expectation $\mu = 20$ and a standard deviation of 2. According to (3.8), there are more than 30 errors with probability

$$\mathbf{P}\{X > 30\} \leq \mathbf{P}\{|X - 20| > 10\} \leq \left(\frac{2}{10}\right)^2 = 0.04.$$

However, if the standard deviation is 5 instead of 2, then the probability of more than 30 errors can only be bounded by $\left(\frac{5}{10}\right)^2 = 0.25$. $\diamond$

Chebyshev's inequality is universal because it works for *any* distribution. Often it gives a rather loose bound for the probability of $|X - \mu| > \varepsilon$. With more information about the distribution, this bound may be improved.

3.3.8 Application to finance

Chebyshev's inequality shows that in general, higher variance implies higher probabilities of *large deviations*, and this increases the *risk* for a random variable to take values far from its expectation.

This finds a number of immediate applications. Here we focus on evaluating risks of financial deals, allocating funds, and constructing optimal portfolios. This application is intuitively simple. The same methods can be used for the optimal allocation of computer memory, CPU time, customer support, or other resources.

Example 3.13 (CONSTRUCTION OF AN OPTIMAL PORTFOLIO). We would like to invest \$10,000 into shares of companies XX and YY. Shares of XX cost \$20 per share. The market analysis shows that their expected return is \$1 per share with a standard deviation of \$0.5. Shares of YY cost \$50 per share, with an expected return of \$2.50 and a standard deviation of \$1 per share, and returns from the two companies are independent. In order to maximize the expected return and minimize the risk (standard deviation or variance), is it better to invest (A) all \$10,000 into XX, (B) all \$10,000 into YY, or (C) \$5,000 in each company?

Solution. Let X be the actual (random) return from each share of XX, and Y be the actual return from each share of YY. Compute the expectation and variance of the return for each of the proposed portfolios (A, B, and C).

(a) At \$20 a piece, we can use \$10,000 to buy 500 shares of XX collecting a profit of $A = 500X$. Using (3.5) and (3.7),

$$\begin{aligned}\mathbf{E}(A) &= 500 \mathbf{E}(X) = (500)(1) = 500; \\ \text{Var}(A) &= 500^2 \text{Var}(X) = 500^2(0.5)^2 = 62,500.\end{aligned}$$

(b) Investing all \$10,000 into YY, we buy $10,000/50=200$ shares of it and collect a profit of $B = 200Y$,

$$\begin{aligned}\mathbf{E}(B) &= 200 \mathbf{E}(Y) = (200)(2.50) = 500; \\ \text{Var}(A) &= 200^2 \text{Var}(Y) = 200^2(1)^2 = 40,000.\end{aligned}$$

(c) Investing \$5,000 into each company makes a portfolio consisting of 250 shares of XX and 100 shares of YY; the profit in this case will be $C = 250X + 100Y$. Following (3.7) for independent X and Y ,

$$\begin{aligned}\mathbf{E}(C) &= 250 \mathbf{E}(X) + 100 \mathbf{E}(Y) = 250 + 250 = 500; \\ \text{Var}(C) &= 250^2 \text{Var}(X) + 100^2 \text{Var}(Y) = 250^2(0.5)^2 + 100^2(1)^2 = 25,625.\end{aligned}$$

Result: The expected return is the same for each of the proposed three portfolios because each share of each company is expected to return $1/20$ or $2.50/50$, which is 5%. In terms of the expected return, all three portfolios are *equivalent*. Portfolio C, where investment is split between two companies, has the lowest variance; therefore, it is the least risky. This supports one of the basic principles in finance: *to minimize the risk, diversify the portfolio*.

◇

Example 3.14 (OPTIMAL PORTFOLIO, CORRELATED RETURNS). Suppose now that the individual stock returns X and Y are no longer independent. If the correlation coefficient is $\rho = 0.4$, how will it change the results of the previous example? What if they are negatively correlated with $\rho = -0.2$?

Solution. Only the volatility of the diversified portfolio C changes due to the correlation coefficient. Now $\text{Cov}(X, Y) = \rho \text{Std}(X) \text{Std}(Y) = (0.4)(0.5)(1) = 0.2$, thus the variance of C increases by $2(250)(100)(0.2) = 10,000$,

$$\begin{aligned} \text{Var}(C) &= \text{Var}(250X + 100Y) \\ &= 250^2 \text{Var}(X) + 100^2 \text{Var}(Y) + 2(250)(100) \text{Cov}(X, Y) \\ &= 25,625 + 10,000 = 35,625. \end{aligned}$$

Nevertheless, the diversified portfolio C is still optimal.

Why did the risk of portfolio C increase due to positive correlation of the two stocks? When X and Y are positively correlated, low values of X are likely to accompany low values of Y ; therefore, the probability of the overall low return is higher, increasing the risk of the portfolio.

Conversely, negative correlation means that low values of X are likely to be compensated by high values of Y , and vice versa. Thus, the risk is reduced. Say, with the given $\rho = -0.2$, we compute $\text{Cov}(X, Y) = \rho \text{Std}(X) \text{Std}(Y) = (-0.2)(0.5)(1) = -0.1$, and

$$\text{Var}(C) = 25,625 + 2(250)(100) \text{Cov}(X, Y) = 25,625 - 5,000 = 20,625.$$

Diversified portfolios consisting of negatively correlated components are the least risky. $\diamond$

Example 3.15 (OPTIMIZING EVEN FURTHER). So, after all, with \$10,000 to invest, what is the most optimal portfolio consisting of shares of XX and YY , given their correlation coefficient of $\rho = -0.2$?

This is an *optimization* problem. Suppose t dollars are invested into XX and $(10,000 - t)$ dollars into YY , with the resulting profit is C_t . This amounts for $t/20$ shares of X and $(10,000 - t)/50 = 200 - t/50$ shares of YY . Plans A and B correspond to $t = 10,000$ and $t = 0$.

The expected return remains constant at \$500. If $\text{Cov}(X, Y) = -0.1$, as in Example 3.14 above, then

$$\begin{aligned} \text{Var}(C_t) &= \text{Var}\{tX + (200 - t/50)Y\} \\ &= (t/20)^2 \text{Var}(X) + (200 - t/50)^2 \text{Var}(Y) + 2(t/20)(200 - t/50) \text{Cov}(X, Y) \\ &= (t/20)^2 (0.5)^2 + (200 - t/50)^2 (1)^2 + 2(t/20)(200 - t/50)(-0.1) \\ &= \frac{49t^2}{40,000} - 10t + 40,000. \end{aligned}$$

See the graph of this function in Figure 3.6. Minimum of this variance is found at $t^* = 10/(\frac{49t^2}{40,000}) = 4081.63$. Thus, for the most optimal portfolio, we should invest \$4081.63 into XX and the remaining \$5919.37 into YY . Then we achieve the smallest possible risk (variance) of $(\$^2)19,592$, measured, as we know, in squared dollars. $\diamond$

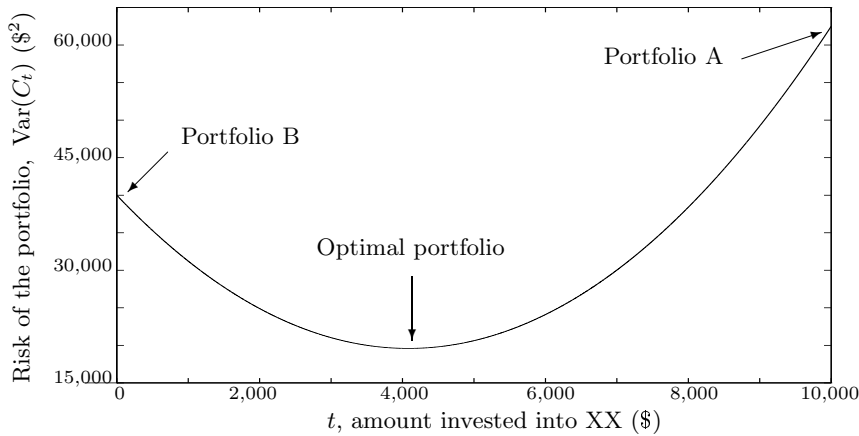


FIGURE 3.6: Variance of a diversified portfolio.

3.4 Families of discrete distributions

Next, we introduce the most commonly used families of discrete distributions. Amazingly, absolutely different phenomena can be adequately described by the same mathematical model, or a family of distributions. For example, as we shall see below, the number of virus attacks, received e-mails, error messages, network blackouts, telephone calls, traffic accidents, earthquakes, and so on can all be modeled by the same Poisson family of distributions.

3.4.1 Bernoulli distribution

The simplest random variable (excluding non-random ones!) takes just two possible values. Call them 0 and 1.

DEFINITION 3.10

A random variable with two possible values, 0 and 1, is called a **Bernoulli variable**, its distribution is **Bernoulli distribution**, and any experiment with a *binary outcome* is called a **Bernoulli trial**.

This distribution is named after a Swiss mathematician *Jacob Bernoulli* (1654-1705) who discovered not only Bernoulli but also Binomial distribution.

Good or defective components, parts that pass or fail tests, transmitted or lost signals, working or malfunctioning hardware, benign or malicious attachments, sites that contain or do not contain a keyword, girls and boys, heads and tails, and so on, are examples of Bernoulli trials. All these experiments fit the same Bernoulli model, where we shall use generic names for the two outcomes: “*successes*” and “*failures*.” These are nothing but

commonly used generic names; in fact, successes do not have to be good, and failures do not have to be bad.

If $P(1) = p$ is the probability of a *success*, then $P(0) = q = 1 - p$ is the probability of a *failure*. We can then compute the expectation and variance as

$$\begin{aligned}\mathbf{E}(X) &= \sum_x P(x) = (0)(1-p) + (1)(p) = p, \\ \text{Var}(X) &= \sum_x (x-p)^2 P(x) = (0-p)^2(1-p) + (1-p)^2 p \\ &= p(1-p)(p+1-p) = p(1-p).\end{aligned}$$

**Bernoulli
distribution**

p	$=$	probability of success
$P(x)$	$=$	$\begin{cases} q = 1 - p & \text{if } x = 0 \\ p & \text{if } x = 1 \end{cases}$
$\mathbf{E}(X)$	$=$	p
$\text{Var}(X)$	$=$	pq

In fact, we see that there is a whole *family of Bernoulli distributions*, indexed by a *parameter* p . Every p between 0 and 1 defines another Bernoulli distribution. The distribution with $p = 0.5$ carries the highest level of uncertainty because $\text{Var}(X) = pq$ is maximized by $p = q = 0.5$. Distributions with lower or higher p have lower variances. Extreme parameters $p = 0$ and $p = 1$ define non-random variables 0 and 1, respectively; their variance is 0.

3.4.2 Binomial distribution

Now consider a sequence of independent Bernoulli trials and count the number of successes in it. This may be the number of defective computers in a shipment, the number of updated files in a folder, the number of girls in a family, the number of e-mails with attachments, etc.

DEFINITION 3.11 —

A variable described as the number of successes in a sequence of independent Bernoulli trials has **Binomial distribution**. Its parameters are n , the number of trials, and p , the probability of success.

Remark: “Binomial” can be translated as “two numbers,” *bi* meaning “two” and *nom* meaning “a number,” thus reflecting the concept of binary outcomes.

Binomial probability mass function is

$$P(x) = \mathbf{P}\{X = x\} = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \dots, n, \quad (3.9)$$

which is the probability of exactly x successes in n trials. In this formula, p^x is the probability of x successes, probabilities being multiplied due to independence of trials. Also, q^{n-x} is

the probability of the remaining $(n - x)$ trials being failures. Finally, $\binom{n}{x} = \frac{n!}{x!(n-x)!}$ is the number of elements of the sample space Ω that form the event $\{X = x\}$. This is the number of possible orderings of x successes and $(n - x)$ failures among n trials, and it is computed as $C(n, x)$ in (2.6).

Due to a somewhat complicated form of (3.9), practitioners use a *table of Binomial distribution*, Table A2. Its entries are values of the Binomial cdf $F(x)$. If we need a pmf instead of a cdf, we can always compute it from the table as

$$P(x) = F(x) - F(x - 1).$$

Example 3.16. As part of a business strategy, randomly selected 20% of new internet service subscribers receive a special promotion from the provider. A group of 10 neighbors signs for the service. What is the probability that at least 4 of them get a special promotion?

Solution. We need to find the probability $P\{X \geq 4\}$, where X is the number of people, out of 10, who receive a special promotion. This is the number of successes in 10 Bernoulli trials, therefore, X has Binomial distribution with parameters $n = 10$ and $p = 0.2$. From Table A2,

$$P\{X \geq 4\} = 1 - F(3) = 1 - 0.8791 = \underline{0.1209}.$$

◇

The table does not go beyond $n = 20$. For large n , we shall learn to use some good approximations.

Let's turn to the Binomial expectation and variance. Computing the expected value directly by (3.3) results in a complicated formula,

$$\mathbf{E}(X) = \sum_{x=0}^n x \binom{n}{x} p^x q^{n-x} = \dots ?$$

A shortcut can be obtained from the following important property.

Each Bernoulli trial is associated with a Bernoulli variable that equals 1 if the trial results in a success and 0 in case of a failure. Then, a sum of these variables is the overall number of successes. Thus, *any Binomial variable X can be represented as a sum of independent Bernoulli variables*,

$$X = X_1 + \dots + X_n.$$

We can use this to compute (referring to (3.5) and (3.7))

$$\mathbf{E}(X) = \mathbf{E}(X_1 + \dots + X_n) = \mathbf{E}(X_1) + \dots + \mathbf{E}(X_n) = p + \dots + p = np$$

and

$$\text{Var}(X) = \text{Var}(X_1 + \dots + X_n) = \text{Var}(X_1) + \dots + \text{Var}(X_n) = npq.$$

**Binomial
distribution**

n	=	number of trials
p	=	probability of success
$P(x)$	=	$\binom{n}{x} p^x q^{n-x}$
$\mathbf{E}(X)$	=	np
$\text{Var}(X)$	=	npq

Example 3.17. An exciting computer game is released. Sixty percent of players complete all the levels. Thirty percent of them will then buy an advanced version of the game. Among 15 users, what is the expected number of people who will buy the advanced version? What is the probability that at least two people will buy it?

Solution. Let X be the number of people (successes), among the mentioned 15 users (trials), who will buy the advanced version of the game. It has Binomial distribution with $n = 15$ trials and the probability of success

$$\begin{aligned} p &= P\{\text{buy advanced}\} \\ &= P\{\text{buy advanced} \mid \text{complete all levels}\} P\{\text{complete all levels}\} \\ &= (0.30)(0.60) = 0.18. \end{aligned}$$

Then we have

$$E(X) = np = (15)(0.18) = \underline{2.7}$$

and

$$P\{X \geq 2\} = 1 - P(0) - P(1) = 1 - (1 - p)^n - np(1 - p)^{n-1} = \underline{0.7813}.$$

The last probability was computed directly by formula (3.9) because the probability of success, 0.18, is not in Table A2. $\diamond$

MATLAB notes

All the distributions discussed in this book are coded in MATLAB Statistics Toolbox. Binomial cdf can be calculated using the MATLAB function `binocdf(x,n,p)`, where n and p are the parameters. Also, `binopdf(x,n,p)` computes the probability mass function, and `binornd(n,p)` generates a Binomial random variable with the given parameters. Alternatively, we can write `cdf('binomial',x,n,p)` for the cdf and `pdf('binomial',x,n,p)` for the pdf.

MATLAB commands for other distributions are similar; see our inventory in Section 12.1.

3.4.3 Geometric distribution

Again, consider a sequence of independent Bernoulli trials. Each trial results in a “success” or a “failure.”

DEFINITION 3.12

The number of Bernoulli trials needed to get the first success has **Geometric distribution**.

Example 3.18. A search engine goes through a list of sites looking for a given key phrase. Suppose the search terminates as soon as the key phrase is found. The number of sites visited is Geometric. $\diamond$

Example 3.19. A hiring manager interviews candidates, one by one, to fill a vacancy. The number of candidates interviewed until one candidate receives an offer has Geometric distribution. $\diamond$

Geometric random variables can take any integer value from 1 to infinity, because one needs at least 1 trial to have the first success, and the number of trials needed is not limited by any specific number. (For example, there is no guarantee that among the first 10 coin tosses there will be at least one head.) The only parameter is p , the probability of a “success.”

Geometric probability mass function has the form

$$P(x) = \mathbf{P} \{ \text{the 1st success occurs on the } x\text{-th trial} \} = (1-p)^{x-1}p, \quad x = 1, 2, \dots,$$

which is the probability of $(x-1)$ failures followed by one success. Comparing with (3.9), there is no number of combinations in this formula because only one outcome has the first success coming on the x -th trial.

This is the first time we see an *unbounded* random variable, that is, with no upper bound. Variable X can take any positive integer value, from 1 to ∞ . It is insightful to check whether $\sum_x P(x) = 1$, as it should hold for all pmf. Indeed,

$$\sum_x P(x) = \sum_{x=1}^{\infty} (1-p)^{x-1}p = \frac{(1-p)^0}{1-(1-p)}p = 1,$$

where we noticed that the sum on the left is a *geometric series* that gave its name to Geometric distribution.

Finally, Geometric distribution has expectation $\mu = 1/p$ and variance $\sigma^2 = (1-p)/p^2$.

PROOF: The geometric series $s(q) = \sum_0^{\infty} q^x$ equals $(1-q)^{-1}$. Taking derivatives with respect to q ,

$$\frac{1}{(1-q)^2} = \left(\frac{1}{1-q} \right)' = s'(q) = \left(\sum_0^{\infty} q^x \right)' = \sum_0^{\infty} xq^{x-1} = \sum_1^{\infty} xq^{x-1}.$$

It follows that for a Geometric variable X with parameter $p = 1 - q$,

$$\mathbf{E}(X) = \sum_{x=1}^{\infty} xq^{x-1}p = \frac{1}{(1-q)^2}p = \frac{1}{p}.$$

Derivation of the variance is similar. One has to take the second derivative of $s(q)$ and get, after a few manipulations, an expression for $\sum x^2 q^{x-1}$. $\square$

**Geometric
distribution**

p	$=$ probability of success	(3.10)
$P(x)$	$= (1-p)^{x-1}p, \quad x = 1, 2, \dots$	
$\mathbf{E}(X)$	$= \frac{1}{p}$	
$\text{Var}(X)$	$= \frac{1-p}{p^2}$	

Example 3.20 (ST. PETERSBURG PARADOX). This paradox was noticed by a Swiss mathematician *Daniel Bernoulli* (1700–1782), a nephew of Jacob. It describes a *gambling strategy that enables one to win any desired amount of money with probability one*.

Isn't it a very attractive strategy? It's real, there is no cheating!

Consider a game that can be played any number of times. Rounds are independent, and each time your winning probability is p . The game does not have to be favorable to you or even fair; this p can be any positive probability. For each round, you bet some amount x . In case of a success, you win x . If you lose a round, you lose x .

The strategy is simple. Your initial bet is the amount that you desire to win eventually. Then, if you win a round, stop. If you lose a round, double your bet and continue.

Let the desired profit be \$100. The game will progress as follows.

Round	Bet	Balance...	
		... if lose	... if win
1	100	−100	+100 and stop
2	200	−300	+100 and stop
3	400	−700	+100 and stop
...	...	...	...

Sooner or later, the game will stop, and at this moment, your balance will be \$100. Guaranteed! However, this is not what D. Bernoulli called a paradox.

How many rounds should be played? Since each round is a Bernoulli trial, the number of them, X , until the first win is a Geometric random variable with parameter p .

Is the game endless? No, on the average, it will last $\mathbf{E}(X) = 1/p$ rounds. In a fair game with $p = 1/2$, one will need 2 rounds, on the average, to win the desired amount. In an “unfair” game, with $p < 1/2$, it will take longer to win, but still a finite number of rounds. For example, if $p = 0.2$, i.e., one win in five rounds, then on the average, one stops after $1/p = 5$ rounds. This is not a paradox yet.

Finally, how much money does one need to have in order to be able to follow this strategy? Let Y be the amount of the last bet. According to the strategy, $Y = 100 \cdot 2^{X-1}$. It is a discrete random variable whose expectation equals

$$\begin{aligned} \mathbf{E}(Y) &= \sum_x (100 \cdot 2^{x-1}) P_X(x) = 100 \sum_{x=1}^{\infty} 2^{x-1} (1-p)^{x-1} p \\ &= 100p \sum_{x=1}^{\infty} (2(1-p))^{x-1} = \begin{cases} \frac{100p}{2(1-p)} & \text{if } p > 1/2 \\ +\infty & \text{if } p \leq 1/2. \end{cases} \end{aligned}$$

This is the St. Petersburg Paradox! A random variable that is always finite has an infinite expectation! Even when the game is fair offering a 50-50 chance to win, one has to be (on the average!) infinitely wealthy to follow this strategy.

To the best of our knowledge, every casino has a limit on the maximum bet, making sure gamblers cannot fully apply this St. Petersburg strategy. When such a limit is enforced, it can be proved that a winning strategy does not exist. $\diamond$

3.4.4 Negative Binomial distribution

When we studied Geometric distribution and St. Petersburg paradox in Section 3.4.3, we played a game until the first win. Now keep playing until we reach k wins. The number of games played is then *Negative Binomial*.

DEFINITION 3.13

In a sequence of independent Bernoulli trials, the number of trials needed to obtain k successes has **Negative Binomial distribution**.

In some sense, Negative Binomial distribution is opposite to Binomial distribution. Binomial variables count the number of successes in a fixed number of trials whereas Negative Binomial variables count the number of trials needed to see a fixed number of successes. Other than this, there is nothing “negative” about this distribution.

Negative Binomial probability mass function is

$$\begin{aligned} P(x) &= \mathbf{P} \{ \text{the } x\text{-th trial results in the } k\text{-th success} \} \\ &= \mathbf{P} \left\{ \begin{array}{l} (k-1) \text{ successes in the first } (x-1) \text{ trials,} \\ \text{and the last trial is a success} \end{array} \right\} \\ &= \binom{x-1}{k-1} (1-p)^{x-k} p^k. \end{aligned}$$

This formula accounts for the probability of k successes, the remaining $(x-k)$ failures, and the number of outcomes—sequences with the k -th success coming on the x -th trial.

Negative Binomial distribution has two parameters, k and p . With $k=1$, it becomes Geometric. Also, each Negative Binomial variable can be represented as a sum of independent Geometric variables,

$$X = X_1 + \dots + X_k, \quad (3.11)$$

with the same probability of success p . Indeed, the number of trials until the k -th success consists of a Geometric number of trials X_1 until the first success, an additional Geometric number of trials X_2 until the second success, etc.

Because of (3.11), we have

$$\begin{aligned} \mathbf{E}(X) &= \mathbf{E}(X_1 + \dots + X_k) = \frac{k}{p}; \\ \text{Var}(X) &= \text{Var}(X_1 + \dots + X_k) = \frac{k(1-p)}{p^2}. \end{aligned}$$

**Negative
Binomial
distribution**

$$\begin{aligned} k &= \text{number of successes} \\ p &= \text{probability of success} \\ P(x) &= \binom{x-1}{k-1} (1-p)^{x-k} p^k, \quad x = k, k+1, \dots \\ \mathbf{E}(X) &= \frac{k}{p} \\ \text{Var}(X) &= \frac{k(1-p)}{p^2} \end{aligned} \quad (3.12)$$

Example 3.21 (SEQUENTIAL TESTING). In a recent production, 5% of certain electronic components are defective. We need to find 12 non-defective components for our 12 new computers. Components are tested until 12 non-defective ones are found. What is the probability that more than 15 components will have to be tested?

Solution. Let X be the number of components tested until 12 non-defective ones are found. It is a number of trials needed to see 12 successes, hence X has Negative Binomial distribution with $k = 12$ and $p = 0.05$.

We need $P\{X > 15\} = \sum_{16}^{\infty} P(x)$ or $1 - F(15)$; however, there is no table of Negative Binomial distribution in the Appendix, and applying the formula for $P(x)$ directly is rather cumbersome. What would be a quick solution?

Virtually any Negative Binomial problem can be solved by a Binomial distribution. Although X is not Binomial at all, the probability $P\{X > 15\}$ can be related to some Binomial variable. In our example,

$$\begin{aligned} P\{X > 15\} &= P\{\text{more than 15 trials needed to get 12 successes}\} \\ &= P\{15 \text{ trials are not sufficient}\} \\ &= P\{\text{there are fewer than 12 successes in 15 trials}\} \\ &= P\{Y < 12\}, \end{aligned}$$

where Y is the number of successes (non-defective components) in 15 trials, which is a Binomial variable with parameters $n = 15$ and $p = 0.95$. From Table A2 on p. 412,

$$P\{X > 15\} = P\{Y < 12\} = P\{Y \leq 11\} = F(11) = \underline{0.0055}.$$

This technique, expressing a probability about one random variable in terms of another random variable, is rather useful. Soon it will help us relate Gamma and Poisson distributions and simplify computations significantly. $\diamond$

3.4.5 Poisson distribution

The next distribution is related to a concept of *rare events*, or Poissonian events. Essentially it means that two such events are extremely unlikely to occur simultaneously or within a very short period of time. Arrivals of jobs, telephone calls, e-mail messages, traffic accidents, network blackouts, virus attacks, errors in software, floods, and earthquakes are examples of rare events. The rigorous definition of rare events is given in Section 6.3.2.

DEFINITION 3.14

The number of rare events occurring within a fixed period of time has **Poisson distribution**.

This distribution bears the name of a famous French mathematician *Siméon-Denis Poisson* (1781–1840).

**Poisson
distribution**

λ	=	frequency, average number of events
$P(x)$	=	$e^{-\lambda} \frac{\lambda^x}{x!}, x = 0, 1, 2, \dots$
$\mathbf{E}(X)$	=	λ
$\text{Var}(X)$	=	λ

PROOF: This probability mass function satisfies $\sum_0^\infty P(x) = 1$ because the Taylor series for $f(\lambda) = e^\lambda$ at $\lambda = 0$ is

$$e^\lambda = \sum_{x=0}^{\infty} \frac{\lambda^x}{x!}, \quad (3.13)$$

and this series converges for all λ .

The expectation $\mathbf{E}(X)$ can be derived from (3.13) as

$$\mathbf{E}(X) = \sum_{x=0}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} = 0 + e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1} \cdot \lambda}{(x-1)!} = e^{-\lambda} \cdot e^\lambda \cdot \lambda = \lambda.$$

Similarly,

$$\mathbf{E}(X^2) - \mathbf{E}(X) = \sum_{x=0}^{\infty} (x^2 - x)P(x) = \sum_{x=2}^{\infty} x(x-1)e^{-\lambda} \frac{\lambda^x}{x!} = e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^{x-2} \cdot \lambda^2}{(x-2)!} = \lambda^2,$$

and therefore,

$$\text{Var}(X) = \mathbf{E}(X^2) - \mathbf{E}^2(X) = (\lambda^2 + \mathbf{E}(X)) - \mathbf{E}^2(X) = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

□

A Poisson variable can take any nonnegative integer value because there may be no rare events within the chosen period, on one end, and the possible number of events is not limited, on the other end. Poisson distribution has one parameter, $\lambda > 0$, which is the average number of the considered rare events. Values of its cdf are given in Table A3 on p. 415.

Example 3.22 (NEW ACCOUNTS). Customers of an internet service provider initiate new accounts at the average rate of 10 accounts per day.

- (a) What is the probability that more than 8 new accounts will be initiated today?
- (b) What is the probability that more than 16 accounts will be initiated within 2 days?

Solution. (a) New account initiations qualify as rare events because no two customers open accounts simultaneously. Then the number X of today's new accounts has Poisson distribution with parameter $\lambda = 10$. From Table A3,

$$\mathbf{P}\{X > 8\} = 1 - F_X(8) = 1 - 0.333 = \underline{0.667}.$$

(b) The number of accounts, Y , opened within 2 days does *not* equal $2X$. Rather, Y is another Poisson random variable whose parameter equals 20. Indeed, the parameter is the average number of rare events, which, over the period of two days, doubles the one-day average. Using Table A3 with $\lambda = 20$,

$$\mathbf{P}\{Y > 16\} = 1 - F_Y(16) = 1 - 0.221 = \underline{0.779}.$$

◇

3.4.6 Poisson approximation of Binomial distribution

Poisson distribution can be effectively used to approximate Binomial probabilities when the number of trials n is large, and the probability of success p is small. Such an approximation is adequate, say, for $n \geq 30$ and $p \leq 0.05$, and it becomes more accurate for larger n .

Example 3.23 (NEW ACCOUNTS, CONTINUED). Indeed, the situation in Example 3.22 can be viewed as a sequence of Bernoulli trials. Suppose there are $n = 400,000$ potential internet users in the area, and on any specific day, each of them opens a new account with probability $p = 0.000025$. We see that the number of new accounts is the number of successes, hence a Binomial model with expectation $\mathbf{E}(X) = np = 10$ is possible. However, a distribution with such extreme n and p is unlikely to be found in any table, and computing its pmf by hand is tedious. Instead, one can use Poisson distribution with the same expectation $\lambda = 10$. $\diamond$

**Poisson
approximation
to Binomial**

$$\text{Binomial}(n, p) \approx \text{Poisson}(\lambda) \\ \text{where } n \geq 30, p \leq 0.05, np = \lambda$$

(3.14)

Remark: Mathematically, it means closeness of Binomial and Poisson pmf,

$$\lim_{\substack{n \rightarrow \infty \\ p \rightarrow 0 \\ np \rightarrow \lambda}} \binom{n}{x} p^x (1-p)^{n-x} = e^{-\lambda} \frac{\lambda^x}{x!}$$

and this is what S. D. Poisson has shown.

When p is large ($p \geq 0.95$), the Poisson approximation is applicable too. The probability of a failure $q = 1 - p$ is small in this case. Then, we can approximate the number of failures, which is also Binomial, by a Poisson distribution.

Example 3.24. Ninety-seven percent of electronic messages are transmitted with no error. What is the probability that out of 200 messages, at least 195 will be transmitted correctly?

Solution. Let X be the number of correctly transmitted messages. It is the number of successes in 200 Bernoulli trials, thus X is Binomial with $n = 200$ and $p = 0.97$. Poisson approximation cannot be applied to X because p is too large. However, the number of failures Y is also Binomial, with parameters $n = 200$ and $q = 0.03$, and it is approximately Poisson with $\lambda = nq = 6$. From Table A3,

$$\mathbf{P}\{X \geq 195\} = \mathbf{P}\{Y \leq 5\} = F_Y(5) \approx \underline{0.446}.$$

$\diamond$

There is a great variety of applications involving a large number of trials with a small probability of success. If the trials are not independent, the number of successes is not Binomial, in general. However, if dependence is weak, the use of Poisson approximation in such problems can still produce remarkably accurate results.

Example 3.25 (BIRTHDAY PROBLEM). This continues Exercise 2.25 on p. 36. Consider a class with $N \geq 10$ students. Compute the probability that at least two of them have their birthdays on the same day. How many students should be in class in order to have this probability above 0.5?

Solution. Poisson approximation will be used for the number of shared birthdays among all

$$n = \binom{N}{2} = \frac{N(N-1)}{2}$$

pairs of students in this class. In each pair, both students are born on the same day with probability $p = 1/365$. Each pair is a Bernoulli trial because the two birthdays either match or don't match. Besides, matches in two different pairs are “nearly” independent. Therefore, X , the number of pairs sharing birthdays, is “almost” Binomial. For $N \geq 10$, $n \geq 45$ is large, and p is small, thus, we shall use Poisson approximation with $\lambda = np = N(N-1)/730$,

$$\mathbf{P}\{\text{there are two students sharing birthday}\} = 1 - \mathbf{P}\{\text{no matches}\}$$

$$= 1 - \mathbf{P}\{X = 0\} \approx 1 - e^{-\lambda} \approx 1 - e^{-N^2/730}.$$

Solving the inequality $1 - e^{-N^2/730} > 0.5$, we obtain $N > \sqrt{730 \ln 2} \approx 22.5$. That is, in a class of at least $N = 23$ students, there is a more than 50% chance that at least two students were born on the same day of the year! $\diamond$

The introduced method can only be applied to very small and very large values of p . For moderate p ($0.05 \leq p \leq 0.95$), the Poisson approximation may not be accurate. These cases are covered by the Central Limit Theorem on p. 93.

Summary and conclusions

Discrete random variables can take a finite or countable number of isolated values with different probabilities. Collection of all such probabilities is a distribution, which describes the behavior of a random variable. Random vectors are sets of random variables; their behavior is described by the joint distribution. Marginal probabilities can be computed from the joint distribution by the Addition Rule.

The average value of a random variable is its expectation. Variability around the expectation is measured by the variance and the standard deviation. Covariance and correlation measure association of two random variables. For any distribution, probabilities of large deviations can be bounded by Chebyshev's inequality, using only the expectation and variance.

Different phenomena can often be described by the same probabilistic model, or a family of distributions. The most commonly used discrete families are Binomial, including Bernoulli, Negative Binomial, including Geometric, and Poisson. Each family of distributions is used for a certain general type of situations; it has its parameters and a clear formula and/or a table for computing probabilities. These families are summarized in Section 12.1.1.

Exercises

3.1. A computer virus is trying to corrupt two files. The first file will be corrupted with probability 0.4. Independently of it, the second file will be corrupted with probability 0.3.

- (a) Compute the probability mass function (pmf) of X , the number of corrupted files.
- (b) Draw a graph of its cumulative distribution function (cdf).

3.2. Every day, the number of network blackouts has a distribution (probability mass function)

x	0	1	2
$P(x)$	0.7	0.2	0.1

A small internet trading company estimates that each network blackout results in a \$500 loss. Compute expectation and variance of this company's daily loss due to blackouts.

3.3. There is one error in one of five blocks of a program. To find the error, we test three randomly selected blocks. Let X be the number of errors in these three blocks. Compute $\mathbf{E}(X)$ and $\text{Var}(X)$.

3.4. Tossing a fair die is an experiment that can result in any integer number from 1 to 6 with equal probabilities. Let X be the number of dots on the top face of a die. Compute $\mathbf{E}(X)$ and $\text{Var}(X)$.

3.5. A software package consists of 12 programs, five of which must be upgraded. If 4 programs are randomly chosen for testing,

- (a) What is the probability that at least two of them must be upgraded?
- (b) What is the expected number of programs, out of the chosen four, that must be upgraded?

3.6. A computer program contains one error. In order to find the error, we split the program into 6 blocks and test two of them, selected at random. Let X be the number of errors in these blocks. Compute $\mathbf{E}(X)$.

3.7. The number of home runs scored by a certain team in one baseball game is a random variable with the distribution

x	0	1	2
$P(x)$	0.4	0.4	0.2

The team plays 2 games. The number of home runs scored in one game is independent of the number of home runs in the other game. Let Y be the *total* number of home runs. Find $\mathbf{E}(Y)$ and $\text{Var}(Y)$.

- 3.8.** A computer user tries to recall her password. She knows it can be one of 4 possible passwords. She tries her passwords until she finds the right one. Let X be the number of wrong passwords she uses before she finds the right one. Find $\mathbf{E}(X)$ and $\text{Var}(X)$.
- 3.9.** It takes an average of 40 seconds to download a certain file, with a standard deviation of 5 seconds. The actual distribution of the download time is unknown. Using Chebyshev's inequality, what can be said about the probability of spending more than 1 minute for this download?
- 3.10.** Every day, the number of traffic accidents has the probability mass function

x	0	1	2	more than 2
$P(x)$	0.6	0.2	0.2	0

independently of other days. What is the probability that there are more accidents on Friday than on Thursday?

- 3.11.** Two dice are tossed. Let X be *the smaller* number of points. Let Y be *the larger* number of points. If both dice show the same number, say, z points, then $X = Y = z$.
- Find the joint probability mass function of (X, Y) .
 - Are X and Y independent? Explain.
 - Find the probability mass function of X .
 - If $X = 2$, what is the probability that $Y = 5$?
- 3.12.** Two random variables, X and Y , have the joint distribution $P(x, y)$,

$P(x, y)$		x	
		0	1
y	0	0.5	0.2
	1	0.2	0.1

- Are X and Y independent? Explain.
 - Are $(X + Y)$ and $(X - Y)$ independent? Explain.
- 3.13.** Two random variables X and Y have the joint distribution, $P(0, 0) = 0.2$, $P(0, 2) = 0.3$, $P(1, 1) = 0.1$, $P(2, 0) = 0.3$, $P(2, 2) = 0.1$, and $P(x, y) = 0$ for all other pairs (x, y) .
- Find the probability mass function of $Z = X + Y$.
 - Find the probability mass function of $U = X - Y$.
 - Find the probability mass function of $V = XY$.
- 3.14.** An internet service provider charges its customers for the time of the internet use rounding

it up to the nearest hour. The joint distribution of the used time (X , hours) and the charge per hour (Y , cents) is given in the table below.

$P(x, y)$		x			
		1	2	3	4
y	1	0	0.06	0.06	0.10
	2	0.10	0.10	0.04	0.04
	3	0.40	0.10	0	0

Each customer is charged $Z = X \cdot Y$ cents, which is the number of hours multiplied by the price of each hour. Find the distribution of Z .

- 3.15.** Let X and Y be the number of hardware failures in two computer labs in a given month. The joint distribution of X and Y is given in the table below.

$P(x, y)$		x		
		0	1	2
y	0	0.52	0.20	0.04
	1	0.14	0.02	0.01
	2	0.06	0.01	0

- (a) Compute the probability of at least one hardware failure.
- (b) From the given distribution, are X and Y independent? Why or why not?
- 3.16.** The number of hardware failures, X , and the number of software failures, Y , on any day in a small computer lab have the joint distribution $P(x, y)$, where $P(0, 0) = 0.6$, $P(0, 1) = 0.1$, $P(1, 0) = 0.1$, $P(1, 1) = 0.2$. Based on this information,
- (a) Are X and Y (hardware and software failures) independent?
- (b) Compute $\mathbf{E}(X + Y)$, i.e., the expected total number of failures during 1 day.
- 3.17.** Shares of company A are sold at \$10 per share. Shares of company B are sold at \$50 per share. According to a market analyst, 1 share of each company can either gain \$1, with probability 0.5, or lose \$1, with probability 0.5, independently of the other company. Which of the following portfolios has the lowest risk:
- (a) 100 shares of A
- (b) 50 shares of A + 10 shares of B
- (c) 40 shares of A + 12 shares of B
- 3.18.** Shares of company A cost \$10 per share and give a profit of $X\%$. Independently of A, shares of company B cost \$50 per share and give a profit of $Y\%$. Deciding how to invest \$1,000, Mr. X chooses between 3 portfolios:

- (a) 100 shares of A,
- (b) 50 shares of A and 10 shares of B,

(c) 20 shares of B.

The distribution of X is given by probabilities:

$$P\{X = -3\} = 0.3, P\{X = 0\} = 0.2, P\{X = 3\} = 0.5.$$

The distribution of Y is given by probabilities:

$$P\{Y = -3\} = 0.4, P\{Y = 3\} = 0.6.$$

Compute expectations and variances of the total dollar profit generated by portfolios (a), (b), and (c). What is the least risky portfolio? What is the most risky portfolio?

3.19. A and B are two competing companies. An investor decides whether to buy

- (a) 100 shares of A, or
- (b) 100 shares of B, or
- (c) 50 shares of A and 50 shares of B.

A profit made on 1 share of A is a random variable X with the distribution $P(X = 2) = P(X = -2) = 0.5$.

A profit made on 1 share of B is a random variable Y with the distribution $P(Y = 4) = 0.2, P(Y = -1) = 0.8$.

If X and Y are independent, compute the expected value and variance of the total profit for strategies (a), (b), and (c).

3.20. A quality control engineer tests the quality of produced computers. Suppose that 5% of computers have defects, and defects occur independently of each other.

- (a) Find the probability of exactly 3 defective computers in a shipment of twenty.
- (b) Find the probability that the engineer has to test at least 5 computers in order to find 2 defective ones.

3.21. A lab network consisting of 20 computers was attacked by a computer virus. This virus enters each computer with probability 0.4, independently of other computers. Find the probability that it entered at least 10 computers.

3.22. Five percent of computer parts produced by a certain supplier are defective. What is the probability that a sample of 16 parts contains more than 3 defective ones?

3.23. Every day, a lecture may be canceled due to inclement weather with probability 0.05. Class cancellations on different days are independent.

- (a) There are 15 classes left this semester. Compute the probability that at least 4 of them get canceled.
- (b) Compute the probability that the tenth class this semester is the third class that gets canceled.

- 3.24.** An internet search engine looks for a certain keyword in a sequence of independent web sites. It is believed that 20% of the sites contain this keyword.
- (a) Compute the probability that at least 5 of the first 10 sites contain the given keyword.
 - (b) Compute the probability that the search engine had to visit at least 5 sites in order to find the first occurrence of a keyword.
- 3.25.** About ten percent of users do not close Windows properly. Suppose that Windows is installed in a public library that is used by random people in a random order.
- (a) On the average, how many users of this computer *do not* close Windows properly before someone *does* close it properly?
 - (b) What is the probability that exactly 8 of the next 10 users will close Windows properly?
- 3.26.** After a computer virus entered the system, a computer manager checks the condition of all important files. She knows that each file has probability 0.2 to be damaged by the virus, independently of other files.
- (a) Compute the probability that at least 5 of the first 20 files are damaged.
 - (b) Compute the probability that the manager has to check at least 6 files in order to find 3 undamaged files.
- 3.27.** Messages arrive at an electronic message center at random times, with an average of 9 messages per hour.
- (a) What is the probability of receiving *at least* five messages during the next hour?
 - (b) What is the probability of receiving *exactly* five messages during the next hour?
- 3.28.** The number of received electronic messages has Poisson distribution with some parameter λ . Using Chebyshev inequality, show that the probability of receiving more than 4λ messages does not exceed $1/(9\lambda)$.
- 3.29.** An insurance company divides its customers into 2 groups. Twenty percent of customers are in the high-risk group, and eighty percent are in the low-risk group. The high-risk customers make an average of 1 accident per year while the low-risk customers make an average of 0.1 accidents per year. Eric had no accidents last year. What is the probability that he is a high-risk driver?
- 3.30.** Eric from Exercise 3.29 continues driving. After three years, he still has no traffic accidents. Now, what is the conditional probability that he is a high-risk driver?
- 3.31.** Before the computer is assembled, its vital component (motherboard) goes through a special inspection. Only 80% of components pass this inspection.

- (a) What is the probability that at least 18 of the next 20 components pass inspection?
- (b) On the average, how many components should be inspected until a component that passes inspection is found?

3.32. On the average, 1 computer in 800 crashes during a severe thunderstorm. A certain company had 4,000 working computers when the area was hit by a severe thunderstorm.

- (a) Compute the probability that less than 10 computers crashed.
- (b) Compute the probability that exactly 10 computers crashed.

You may want to use a suitable approximation.

3.33. The number of computer shutdowns during any month has a Poisson distribution, averaging 0.25 shutdowns per month.

- (a) What is the probability of at least 3 computer shutdowns during the next year?
- (b) During the next year, what is the probability of at least 3 months (out of 12) with exactly 1 computer shutdown in each?

3.34. A dangerous computer virus attacks a folder consisting of 250 files. Files are affected by the virus independently of one another. Each file is affected with the probability 0.032. What is the probability that more than 7 files are affected by this virus?

3.35. In some city, the probability of a thunderstorm on any day is 0.6. During a thunderstorm, the number of traffic accidents has Poisson distribution with parameter 10. Otherwise, the number of traffic accidents has Poisson distribution with parameter 4. If there were 7 accidents yesterday, what is the probability that there was a thunderstorm?

3.36. An interactive system consists of ten terminals that are connected to the central computer. At any time, each terminal is ready to transmit a message with probability 0.7, independently of other terminals. Find the probability that exactly 6 terminals are ready to transmit at 8 o'clock.

3.37. Network breakdowns are unexpected rare events that occur every 3 weeks, on the average. Compute the probability of more than 4 breakdowns during a 21-week period.

3.38. Simplifying expressions, derive from the definitions of variance and covariance that

- (a) $\text{Var}(X) = \mathbf{E}(X^2) - \mathbf{E}^2(X)$;
- (b) $\text{Cov}(X, Y) = \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y)$.

3.39. Show that

$$\begin{aligned} & \text{Cov}(aX + bY + c, dZ + eW + f) \\ &= ad \text{Cov}(X, Z) + ae \text{Cov}(X, W) + bd \text{Cov}(Y, Z) + be \text{Cov}(Y, W) \end{aligned}$$

for any random variables X, Y, Z, W , and any non-random numbers a, b, c, d, e, f .

